Analogue Electronics II (E2-2) — SOLUTION

① (a) (iii) Circuit shown is a "digital" inverter. Should identify that this is in fact two CS amps. (1)

$\therefore A_V = -G_m R_{out}$ where $G_m = g_{m1} + g_{m2}$

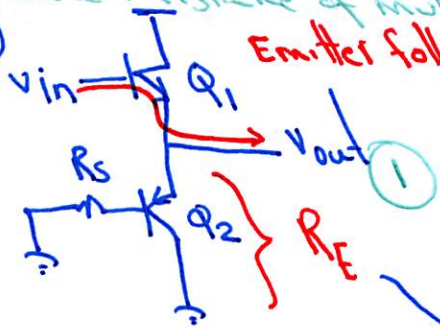
(2) or $G_m = 2g_{m1}$ (assuming g_m's are matched)

$R_{out} = r_{o1} \| r_{o2}$ if both devices are in sat.

$\therefore A_V = -2g_{m1}(r_{o1} \| r_{o2})$ (2)

most students realised this was 2 CS amplifiers in parallel but several made mistake of multiplying g_ms instead of adding

(ii)



(1)

from formula sheet

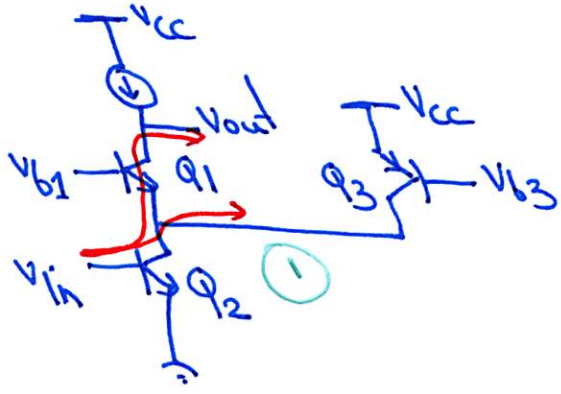
$$A_V = \frac{R_E \| r_o}{\frac{1}{g_m} + R_E \| r_o}$$ (1)

from formula sheet

$$R_E = \frac{1}{g_m} + \frac{R_S}{\beta + 1}$$ (1)

$$\therefore A_V = \frac{\left(\frac{1}{g_{m2}} + \frac{R_S}{\beta+1}\right) \| r_{o1}}{\left(\frac{1}{g_{m2}} + \frac{R_S}{\beta+1}\right) \| r_{o1} + \frac{1}{g_{m1}}}$$ (2)

(iii)



(1)

use $A_V = -G_m R_{out}$

for $G_m \rightarrow v_{in} = v_{be2}$

$i_{out} \neq i_{c2}$

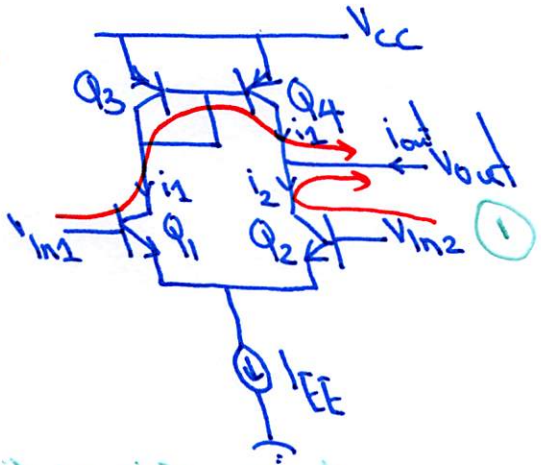
$\therefore$ need $\frac{i_{out}}{i_{c2}}$ → current splitter

$$G_m = \frac{r_{o2} \| r_{o3}}{r_{o2} \| r_{o3} + 1/g_{m1}} (g_{m2})$$ (2)

$$R_{out} = g_{m1} r_{o1} (r_{o2} \| r_{o3} \| r_{\pi 1}) + r_{o1} + (r_{o2} \| r_{o3} \| r_{\pi 1})$$

(2)

(iv)



(1)

use $A_v = -G_m R_{out}$

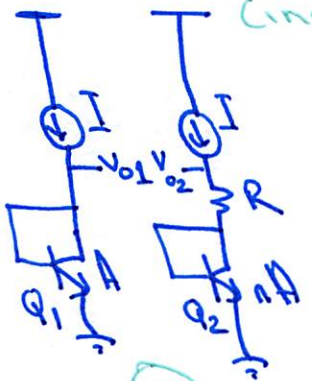
$G_m = g_{m1}$ (if $g_{m1} = g_{m2}$ and $v_{in} = V_{in2} - V_{in1}$) (2)

$R_{out} = r_{o2} \| r_{o4}$

$\therefore A_v = -g_{m1}(r_{o2} \| r_{o4})$ (2)

(ii) to (iv) generally answered well. In (iv) some students applied (incorrectly) ½ ckt assumption)

(b)



(2)

Bandgap voltage reference uses the fact that $V_T \left(= \frac{kT}{q}\right)$ has a +ve TC and V_{BE} has a −ve TC in order to generate a voltage that is independant of temperature. (2)

On ckt on left if $V_{o1} = V_{o2}$

$V_{BE1} = IR + V_{BE2}$ then $IR = V_{BE1} - V_{BE2}$

$= V_T \ln(n)$

$V_{o2} = V_{BE2} + V_T \ln(n)$ (1)

→ need to design n.

(c)

(i) $CMRR = \frac{A_{VDM}}{A_{VCM}}$ (1)

- A_{VDM} is $-G_m R_{out}$ so increasing bias current or o/p R will improve (2)
- A_{VCM} is affected by how good current source the bias sink is (ideally ∞ r_{out}). (2)

$\therefore$ so transconductance (or bias current magnitude) affect A_{VDM} whereas current sink r_o affects A_{VCM}

answered well

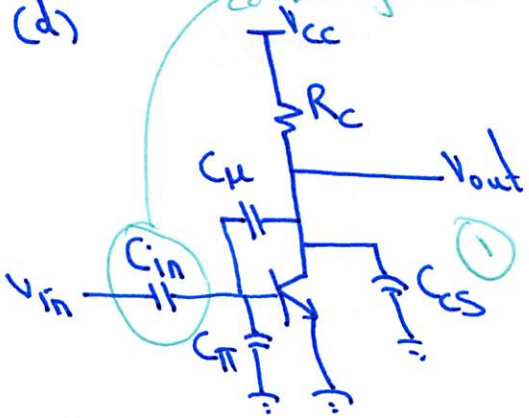
(ii) input offset voltage has two components (systematic offset and random offset). (1)

Systematic offset is affected by an unbalanced o/p i.e ideally $V_{out} = A_0(V_+ - V_-)$ so if $(V_+ - V_-) = 0$ $\therefore V_{out} = 0$. (2)

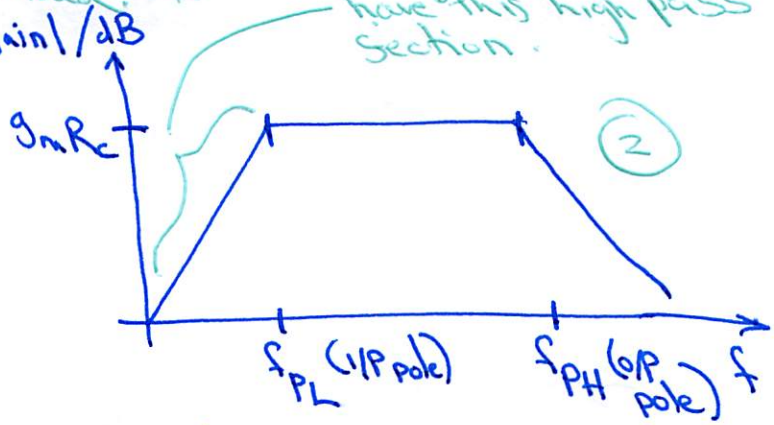
Random offset is due to mismatch between diff. pair. (2)

only a few students did this

(d)



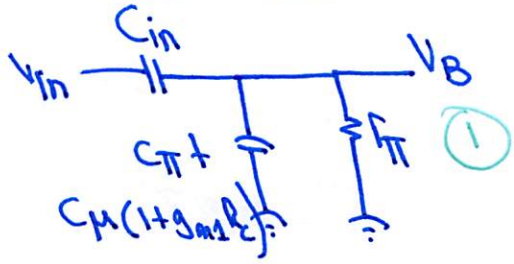
Cin was in many cases not series coupling but parallel load.



many students did not have this high pass section.

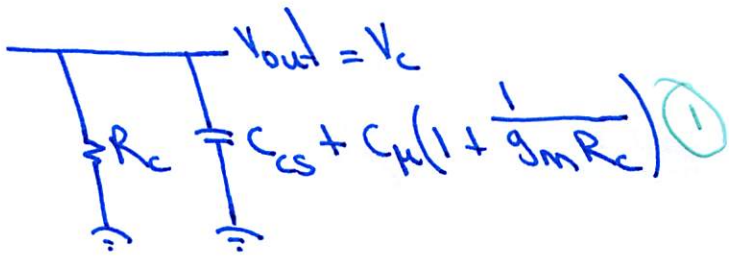
(2)

At input



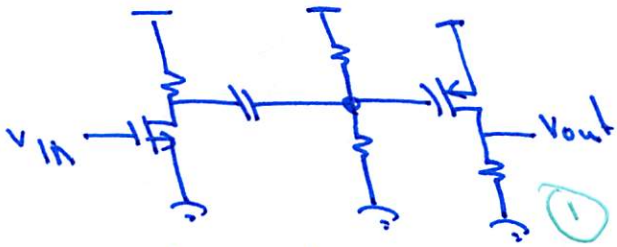
(1)

At output



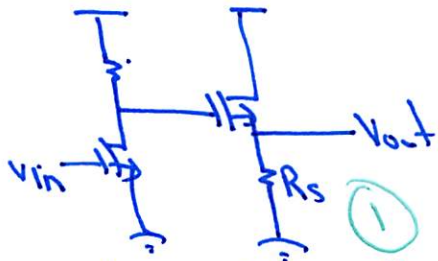
(1)

(e)



(1)

(Capacitive coupling)



(1)

(Direct coupling)

In capacitive coupling each amplifier stage is independently biased and subsequent stages are cascaded using an AC coupling capacitor. (1)

Direct coupling is when amplifier stages are directly cascaded and thus biasing is all dependant. (1)

In IC design large capacitors are unfeasible ∴ capacitive coupling is undesirable for LF signals. (1)

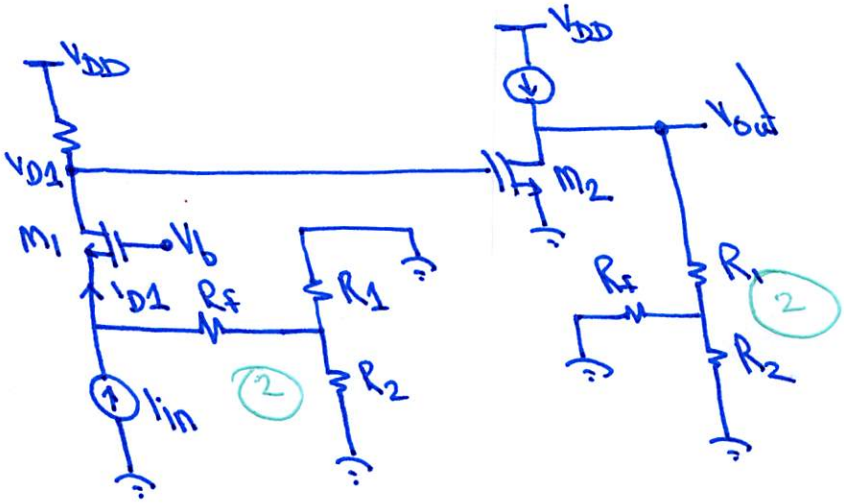
(e) answered well

(2) (a) Since input = current, output = voltage } ∴ Transimpedance amplifier (units of gain = Ohms) (1)

almost all got this.

Ideally $R_{in} = 0$, $R_{out} = 0$. (1) (1)

(b) (i) To determine open loop gain, i.e $\left(\frac{V_{out}}{I_{in}}\right)_{OL}$ need to break feedback loop.



$$A_{OL} = \frac{V_{out}}{I_{in}} = \frac{I_{D1}}{I_{in}} \times \frac{V_{D1}}{I_{D1}} \times \frac{V_{out}}{V_{D1}}$$

$$\frac{I_{D1}}{I_{in}} = \frac{(R_f + R_1 \| R_2)}{(R_f + R_1 \| R_2) + 1/g_{m1}}$$ (1)

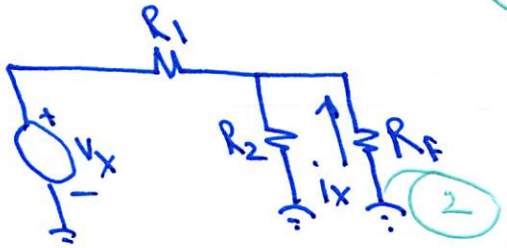
several students assumed $I_{in} = I_{D1}$

$$V_{D1} = I_{D1} R_{D1}$$ (1)

$$\frac{V_{out}}{V_{D1}} = -g_{m2}\left(r_{o2} \| (R_1 + R_f \| R_2)\right)$$ (1)

$$\therefore A_{OL} = -g_{m2}\left[r_{o2} \| (R_1 + R_2 \| R_f)\right] R_{D1} \frac{R_f + R_1 \| R_2}{\frac{1}{g_{m1}} + R_f + R_1 \| R_2}$$ (3)

(ii)



$$K = \frac{I_x}{V_x}$$

generally macromodel provided by most

$$K = \frac{i_x}{v_x} = \frac{-R_2}{(R_1 + R_2 \| R_f)(R_2 + R_f)} = \frac{-R_2 \| R_f}{R_f(R_1 + R_2 \| R_f)}$$ (3)

about 1/2 students got this

(iii) $R_{in,open} = \frac{1}{g_{m1}} \| (R_f + R_1 \| R_2)$ (3)

$R_{out,open} = r_{o2} \| (R_1 + R_2 \| R_f)$ (2)

answered very well

(iv) $$\frac{V_{out}}{I_{in}} = - \frac{g_{m2}[r_{o2} \| (R_1 + R_2 \| R_f)] R_{D1} \frac{R_f + R_1 \| R_2}{\frac{1}{g_{m1}} + R_f + R_1 \| R_2}}{1 + \left\{ g_{m2}[r_{o2} \| (R_1 + R_2 \| R_f)] R_{D1} \frac{R_f + R_1 \| R_2}{\frac{1}{g_{m1}} + R_f + R_1 \| R_2} \right\}}$$ (3)

$$\left\{ \frac{R_2 \| R_f}{R_f(R_1 + R_2 \| R_f)} \right\}$$

$$R_{in,closed} = \frac{\frac{1}{g_{m1}} \| (R_f + R_1 \| R_2)}{1 + A_{OL} K}$$ (2)

$$R_{out,closed} = \frac{r_{o2} \| (R_1 + R_2 \| R_f)}{1 + A_{OL} K}$$ (2)

$1 + A_{OL} K =$

few student got complete, correct expressions

(3)(a) $P_{total} = V_{DD} \times I_{total}$

$I_{total} = I_{ref} + I_{D7} + I_{D8} + I_{D9}$ (2)

From $\frac{W}{L}$'s ratio $(I_{ref} : I_{D7} : I_{D8} : I_{D9}) = 1 : 4 : 8 : 40$

$\therefore I_{total} = I_{ref}(1+4+8+40) = 0.25\,mA(53) = 13.25\,mA$

$\therefore P_{total} = 23.85\,mW$ (2) ← most obtained correct value.

(b) $A_V = \frac{V_{out}}{V_{in2} - V_{in1}} = \frac{V_{out}}{V_z} \times \frac{V_z}{V_y} \times \frac{V_y}{V_{in2} - V_{in1}}$

$\frac{V_y}{V_{in2} - V_{in1}} = -g_{m1}(r_{o2} \| r_{o4})$ (assuming good matching)

$$\frac{v_z}{v_y} = -g_{m5}(r_{o5} \| r_{o8})$$

$$\frac{v_{out}}{v_z} = \frac{r_{o6} \| r_{o9}}{r_{o6} \| r_{o9} + 1/g_{m6}}$$

answered very well

$$\therefore A_v = g_{m1} g_{m5} (r_{o2} \| r_{o4})(r_{o5} \| r_{o8}) \frac{r_{o6} \| r_{o9}}{r_{o6} \| r_{o9} + 1/g_{m6}}$$ (5)

$$g_{m1} = \sqrt{2 \mu_n C_{ox} \left(\frac{W}{L}\right)_1 I_{D1}} = \sqrt{2(200\mu)(250)(0.5m)} = 7.07mS$$

$$g_{m5} = \sqrt{2(100\mu)50(2m)} = 4.47mS$$

$$g_{m6} = \sqrt{2(200\mu)50(10m)} = 14.14mS$$

$$r_{o2} = \frac{1}{\lambda_n I_{D2}} = \frac{1}{(0.1)(0.5m)} = 20k\Omega$$

$$r_{o4} = \frac{1}{(0.2)(0.5m)} = 10k\Omega$$

$$r_{o5} = \frac{1}{(0.2)(2m)} = 2.5k\Omega \qquad r_{o6} = \frac{1}{(0.1)(10m)} = 1k\Omega$$

$$r_{o8} = \frac{1}{(0.1)(2m)} = 5k\Omega \qquad r_{o9} = \frac{1}{(0.1)(10m)} = 1k\Omega$$

(3) Some rounding errors or incorrect expression but generally most followed correct method

$$A_v = (7.07m)(4.47m)(20k \| 10k)(2.5k \| 5k) \frac{1k \| 1k}{1k \| 1k + \frac{1}{14.14m}}$$

31.60μ $\quad$ $6.6k$ $\quad$ $1.6k$ $\quad$ 0.88

$$= 293.65 = 49.36dB$$ (2)

(c) $R_{out} = \frac{1}{g_{m6}} \| r_{o6} \| r_{o9} = \frac{1}{14.14m} \| 1k \| 1k = 61.96\Omega$ (2)

(d) Nodes in signal path : $V_{in1}, V_{in2}, V_x, V_y, V_{out}, V_z$

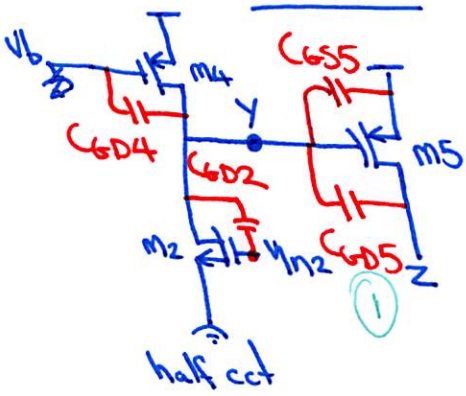
However can exclude low Z nodes (source connections) such as V_x, V_{out}. (1)

$\therefore$ Remaining high Z nodes V_y and V_z (1)

many incorrectly identified V_{out} as high Z

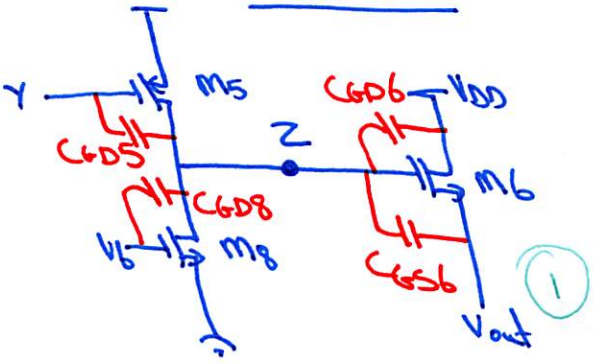
(e) Need to determine RCs at nodes Y and Z.

Node Y



(1)

Node Z



(1)

$R_y = r_{O4} || r_{O2} = 10k\Omega || 20k\Omega$
$= 6.6k\Omega$ (1)

$R_z = r_{O5} || r_{O8} = 2.5k\Omega || 5k\Omega$
$= 1.6k\Omega$ (1)

$C_y = C_{GD4} + C_{GD2}\left(1 - \frac{1}{A_{V2}}\right)$
$+ C_{GS5} + C_{GD5}(1 - A_{V5})$
$= 0.5f(5) + \overset{0.5f}{\cancel{2f}}(250)(21)$
$+ 2f(50)(1) + 0.5f(50)(1+7.45)$
$= 1.34pf$ (2)

$C_z = C_{GD5}\left(1 - \frac{1}{A_{V5}}\right) + C_{GD8}$
$+ C_{GD6} + C_{GS6}(1 - A_{V6})$
$= 0.5f(50)\left(1 + \frac{1}{\cancel{7.45}}\right) + 0.5f(16)$
$+ 0.5f(50) + \cancel{2}0f(50)(1-0.88)$
$= 178ff$ (2)

$f_{Py} = \frac{1}{2\pi R_y C_y} = 17.82MHz$ (1)

$f_{Pz} = \frac{1}{2\pi R_z C_z} = 536.69MHz$ (1)

Although method followed correctly by many, either due to selecting wrong nodes, or incorrect calculation of capacitances, final pole frequencies were incorrect.

Most students however received credit for correct method.